

DATA STRUCTURES HOMEWORK - 2

1. Define an integer array. Declare an integer pointer and initialize it to point to the first element of the array. Use pointer arithmetic to traverse the array and print the elements. Modify the elements of the array using pointer arithmetic.

Code:

```
#include <stdio.h>
int main(){
    int arr[100],n,i,m=30;
    int *p=arr;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter values:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("Traversing through the array:");
    for(i=0;i<n;i++){
        printf("%d ",*(p+i));
    }
    printf("\nModifying the array:");
    for(i=0;i<n;i++){
        *(p+i)=m+i;
    }
    for(i=0;i<n;i++){
        printf("%d ",*(p+i));
    }
    return 0;
}
```

Output:

```
PS D:\ds_hw_2> ./arithi
Enter the number of elements:5
Enter values:12
32
15
67
56
Traversing through the array:12 32 15 67 56
Modifying the array:30 31 32 33 34
PS D:\ds_hw_2>
```

2. Calculate the string length using pointers.

Code:

```
#include <stdio.h>
int main(){
    char arr[100],*p=arr;
    int length=0;
```

```

printf("Enter the string:");
scanf("%s",arr);
while(*p!='\0'){
    length++;
    p++;
}
printf("The length of the array is %d",length);
return 0;
}

```

Output:

```

Modifying the array:30 31 32 33 34
● PS D:\ds_hw_2> gcc strlen.c -o strlen
● PS D:\ds_hw_2> ./strlen
Enter the string:Shobii
The length of the array is 6
○ PS D:\ds_hw_2> █

```

3. Write a function that calculates both the sum and product of two numbers. Return these two results to the function pointer passed as arguments to the function.

Code:

```

#include <stdio.h>
void calculate(int a, int b, int *sum, int *product)
{
    *sum = a + b;
    *product = a * b;
}
int main(){
    int a,b;
    int sum, product;
    printf("Enter Two numbers:");
    scanf("%d %d",&a,&b);
    calculate(a, b, &sum, &product);
    printf("Sum = %d\n", sum);
    printf("Product = %d\n", product);
    return 0;
}

```

Output:

```

● PS D:\ds_hw_2> gcc return.c -o return
● PS D:\ds_hw_2> ./return
Enter Two numbers:56
34
Sum = 90
Product = 1904
○ PS D:\ds_hw_2> █

```

4. Create an array of structures. Use dynamic memory allocation to allocate memory space and use a structure pointer to access, modify, and store the data.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main(){
    int *arr,n,i;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    arr=(int *)malloc(n*sizeof(int));
    printf("Enter the elements:");
    for(i=0;i<n;i++){
        scanf("%d",arr+i);
    }
    for(i=0;i<n;i++){
        printf("%d ",*(arr+i));
    }
    printf("\nModification of the array:");
    for(i=0;i<n;i++){
        printf("%d ",(*(arr+i))*2);
    }
    return 0;
}
```

Output:

```
● PS D:\ds_hw_2> gcc dynamic.c -o dynamic
● PS D:\ds_hw_2> ./dynamic
Enter the number of elements:5
Enter the elements:23
45
32
12
67
23 45 32 12 67
Modification of the array:46 90 64 24 134
○ PS D:\ds_hw_2> █
```

5. Define two functions with the same signature. Declare a function pointer and use the pointer to call both functions.

Code:

```
#include <stdio.h>
int add(int x,int y){
    return x+y;
}
int div(int x,int y){
    if(y==0){
        printf("\nDivision is not possible");
        return 0;
    }
    return x/y;
}
int main(){
```

```

int (*ptr)(int,int);
int a,b;
printf("Enter two numbers:");
scanf("%d %d",&a,&b);
ptr=add;
printf("Addition of two numbers: %d",ptr(a,b));
ptr=div;
printf("\nDivision of two numbers: %d",ptr(a,b));
return 0;
}

```

Output:

```

Division of two numbers: 0
● PS D:\ds_hw_2> gcc sign.c -o sign
● PS D:\ds_hw_2> ./sign
Enter two numbers:80
4
Addition of two numbers: 84
Division of two numbers: 20
○ PS D:\ds_hw_2> █

```

6. Declare an integer, a pointer to it, and a double pointer. Print the values and addresses at each level.

Code:

```

#include <stdio.h>
int main(){
    int a,*p;
    int **pd;
    p=&a;
    pd=&p;
    printf("Enter the value:");
    scanf("%d",&a);
    printf("Value of the number: %d",a);
    printf("\nValue stored in the pointer %ld",p);
    printf("\nValue stored in the double pointer %ld",pd);
    printf("\n*p= %d",*p);
    printf("\n*pd= %d",*pd);
    printf("\n**pd= %d",**pd);
    return 0;
}

```

Output:

```

● PS D:\ds_hw_2> gcc double.c -o double
● PS D:\ds_hw_2> ./double
Enter the value:73
Value of the number: 73
Value stored in the pointer 1491074644
Value stored in the double pointer 1491074632
*p= 73
*pd= 1491074644
**pd= 73
○ PS D:\ds_hw_2> █

```

7. Sort an array of floating-point numbers using Selection Sort.

Code:

```
#include <stdio.h>

int main(){
    float arr[100];int min;
    int i,n,j,k;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter the elements:");
    for(i=0;i<n;i++){
        scanf("%f",&arr[i]);
    }
    for(i=0;i<n;i++){
        min=i;
        for(j=i+1;j<n;j++){
            if(arr[j]<arr[i]){
                min=j;
            }
        }
        float temp;
        temp=arr[i];
        arr[i]=arr[min];
        arr[min]=temp;
    }
    printf("\nSorted array:");
    for(i=0;i<n;i++)
        printf("%.2f ",arr[i]);
    return 0;
}
```

Output:

```
PS D:\ds_hw_2> gcc sele.c -o sele
PS D:\ds_hw_2> ./sele
Enter the number of elements:4
Enter the elements:12.34
45.6
78.9
4.0

Sorted array:4.00 12.34 45.60 78.90
PS D:\ds_hw_2> █
```

8. Define a structure student with id,name, and grade. Create an array of student structures and sort them based on different fields.

Code:

```
#include <stdio.h>
#include <string.h>

struct Student{
    int id;
```

```

    char name[100];
    float grade;
};

int main(){
    struct Student s[10],d[100],a[100],g[100],temp;
    int n,i,j,min;
    printf("Enter the number of students:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("Enter ID:");
        scanf("%d",&s[i].id);
        printf("Enter name: ");
        scanf("%s",s[i].name);
        printf("Enter grade obtained: ");
        scanf("%f",&s[i].grade);
    }
    for(i=0;i<n-1;i++){
        for(j=i+1;j<n;j++){
            if(s[i].id>s[j].id){
                temp.id=s[i].id;
                s[i].id=s[j].id;
                s[j].id=temp.id;
                strcpy(temp.name,s[i].name);
                strcpy(s[i].name,s[j].name);
                strcpy(s[j].name,temp.name);
                temp.grade=s[i].grade;
                s[i].grade=s[j].grade;
                s[j].grade=temp.grade;
            }
        }
    }
    printf("\nSorted based on ID:");
    for(i=0;i<n;i++){
        printf("\nName: %s",s[i].name);
        printf("\nRollno: %d",s[i].id);
        printf("\nGrade: %f",s[i].grade);
    }

    return 0;
}

```

Output:

```
• PS D:\ds_hw_2> gcc struct.c -o struct
• PS D:\ds_hw_2> ./struct
Enter the number of students:3
Enter ID:100
Enter name: Shobi
Enter grade obtained: 100
Enter ID:98
Enter name: Naagul
Enter grade obtained: 99
Enter ID:67
Enter name: Moni
Enter grade obtained: 56

Sorted based on ID:
Name: Moni
Rollno: 67
Grade: 56.000000
Name: Naagul
Rollno: 98
Grade: 99.000000
Name: Shobi
Rollno: 100
Grade: 100.000000
• PS D:\ds_hw_2> █
```

9. Sort a set of negative numbers using Radix Sort.

Code:

```
#include <stdio.h>

void radix(int arr[],int n){
    int i,s,max,min,div=1,j,k,remain,bucket[10][100],bucketcount[10];
    max=arr[0];
    min=arr[0];
    for(i=0;i<n;i++){
        if(min>arr[i]){
            min=arr[i];
        }
    }
    min=-min;
    for(i=0;i<n;i++){
        arr[i]=arr[i]+min;
    }
    for(i=0;i<n;i++){
        if(max<arr[i]){
            max=arr[i];
        }
    }
    while(max/div>0){
        for(i=0;i<10;i++){
            bucketcount[i]=0;
        }
        for(i=0;i<n;i++){
            remain=(arr[i]/div)%10;
```

```

        bucket[remain][bucketcount[remain]]=arr[i];
        bucketcount[remain]++;
    }
    k=0;
    for(i=0;i<10;i++){
        for(j=0;j<bucketcount[i];j++){
            arr[k++]=bucket[i][j];
        }
    }
    div*=10;
}
for(i=0;i<n;i++){
    arr[i]=arr[i]-min;
}
printf("Sorted array:");
for(i=0;i<n;i++){
    printf("%d ",arr[i]);
}
}

int main(){
    int arr[100],n,i;
    printf("Enter the no of elements:");
    scanf("%d",&n);
    printf("Enter the elements:");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    radix(arr,n);
    return 0;
}

```

Output:

```

Sorted array: -12 -25 -15 -78
● PS D:\ds_hw_2> gcc negative.c -o negative
● PS D:\ds_hw_2> ./negative
Enter the no of elements:4
Enter the elements:-10
-9
-11
-89
Sorted array:-89 -11 -10 -9
○ PS D:\ds_hw_2>

```

10. Write a C program to copy an array to another using pointers.

Code:

```
#include <stdio.h>

int main(){
    int arr[100],arr1[100],i,n;
    int *p,*q;
    p=arr;
    q=arr1;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        scanf("%d",p+i);
    }
    for(i=0;i<n;i++){
        *(q+i)=*(p+i);
    }
    printf("\nCopied Array:");
    for(i=0;i<n;i++){
        printf("%d ",*(q+i));
    }
    return 0;
}
```

Output:

```
Sorted array: 89 11 10 5
● PS D:\ds_hw_2> gcc copy.c -o copy
● PS D:\ds_hw_2> ./copy
Enter the number of elements:3
12
100
189

Copied Array:12 100 189
● PS D:\ds_hw_2> █
```